

CO-3 AT-2

17. MUSIC Algorithm Simulation

Simulate the MUSIC algorithm to resolve two closely spaced sinusoidal frequencies in the presence of noise. (25) (BL4) (WK4)

Given Data:

- Frequencies = 500 Hz and 520 Hz
- Sampling Frequency = 5 kHz
- SNR = 10 dB

Tasks:

1. Generate the signal.
2. Form covariance matrix.
3. Perform eigenvalue decomposition.
4. Implement MUSIC algorithm.
5. Estimate frequencies.
6. Compare with periodogram results.

PROGRAM

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Step 1: Generate the Signal
```

```
Fs = 5000;      % Sampling Frequency (Hz)
```

```
f1 = 500;       % Frequency 1
```

```
f2 = 520;       % Frequency 2
```

```
N = 1024;      % Number of samples
SNR = 10;      % Signal-to-Noise Ratio (dB)
```

```
t = (0:N-1)/Fs;
```

```
% Two sinusoidal signals
x = sin(2*pi*f1*t) + sin(2*pi*f2*t);
```

```
% Add White Gaussian Noise
x = awgn(x,SNR,'measured');
```

```
%% Step 2: Form Covariance Matrix
```

```
M = 20;        % Number of array elements (subspace dimension)
```

```
X = zeros(M,N-M+1);
```

```
for i = 1:N-M+1
    X(:,i) = x(i:i+M-1)';
end
```

```
% Covariance Matrix
R = (X*X')/(N-M+1);
```

```
%% Step 3: Eigenvalue Decomposition
```

```
[E,D] = eig(R);
```

```
% Sort Eigenvalues in descending order
```

```
[eigenvalues,index] = sort(diag(D),'descend');
```

```
E = E(:,index);
```

```
% Number of signal sources
```

```
P = 2;
```

```
% Noise Subspace
```

```
En = E(:,P+1:end);
```

```
%% Step 4: MUSIC Algorithm
```

```
f = 0:0.1:1000;      % Frequency search range
```

```
Pmusic = zeros(length(f),1);
```

```
for k = 1:length(f)
```

```
    a = exp(-1j*2*pi*(0:M-1)*f(k)/Fs);
```

```
    Pmusic(k) = 1/(a'*En*En'*a);
```

```
end
```

```
Pmusic = abs(Pmusic);
```

```
Pmusic = Pmusic/max(Pmusic);
```

```
%% Step 5: Estimate Frequencies
```

```
[pks,locs] = findpeaks(Pmusic,f,'SortStr','descend');
```

```
Estimated_Frequencies = sort(locs(1:2));
```

```
disp('Estimated Frequencies using MUSIC (Hz)');
```

```
disp(Estimated_Frequencies);
```

```
%% Step 6: Periodogram
```

```
[pxx,freq] = periodogram(x,[],4096,Fs);
```

```
%% Plot Results
```

```
figure;
```

```
subplot(2,1,1)
```

```
plot(freq,10*log10(pxx),'LineWidth',1.5)
```

```
grid on
```

```
xlabel('Frequency (Hz)')
```

```
ylabel('Power (dB)')
```

```
title('Periodogram')
```

```
subplot(2,1,2)
```

```
plot(f,10*log10(Pmusic),'r','LineWidth',1.5)
```

```
grid on
```

```
xlabel('Frequency (Hz)')
```

```
ylabel('Pseudo Spectrum (dB)')
```

```
title('MUSIC Spectrum')
```

OUTPUT

Estimated Frequencies using MUSIC (Hz)

140.2000 509.6000

